EPSR-test-ex42 writexyz: append and numbered-file modes

Program(s): writexyz | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

Exercises `writexyz`, which exports (part of) an EPSR configuration [1] as `.xyz` blocks for visualisation: optionally centred on one molecule (minimum image), rotated by three Euler angles, restricted to a rectangular slab, with atom types excluded, either appended to `<file>append.xyz` or written as numbered files with a block counter.

Code exercised

Fortran (original, fortran_src/):

- `writexyz.f`: `run_writexyz` (command-line words / stdin prompts), `writexyz` (selection, rotation, block counting, append / numbered files)
- `ato_read_and_save.f`: `minitc` (reading the `.ato`); `minimagec.f`: `minimage` (minimum image about the centre molecule)
- `ato_manipulation_rev4.f`: `centreofmass`, `rotatecoords` (Euler rotation)

Rust (port, rust/epsr-core/src/):

- `tools/writexyz.rs`: `run`, `writexyz`, `block_atoms`
- `ato.rs`: `State::minitc`; `neighbour.rs`: `minimage`
- `tools/groupc_common.rs`: `centreofmass`, `rotatecoords`

Physical system

The EPSR-test configuration `200DMSO400Water.ato`: 200 dimethyl sulphoxide and 400 water molecules (3200 atoms) at 300 K in a cubic box of side 33.1213 Å, and the single-molecule template `DMSO.ato` (10 atoms). No new simulation; the coordinates are only transformed: positions relative to molecule 5 (minimum image), rotated by (phi, theta, chi) = (30°, 45°, 60°), kept if |x|, |y| <= 8 Å and -6 <= z <= 6 Å, with DMSO hydrogen (type 4) removed.

Inputs

EPSR-test files and `writexyz_dmso.stdin`. Two identical slab runs (`append 5 8 8 -6 6 30 45 60 4`), two whole-box runs (`noappend 0 100 100 -100 100 0 0 0 0`), and `writexyz DMSO.ato` with every parameter from stdin (append y, centre 1, distances over two lines, angles 10 20 30, exclusion `3 4`).

How the accuracy was verified

`commands.txt` (5 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

Files compared: `200DMSO400Waterappend.xyz` (two blocks of 179 atoms; the second run reads the existing block and appends block 00002), `200DMSO400Water00001.xyz` and `00002.xyz` (all 3200 atoms), the counter file `200DMSO400Water.nblock`, and `DMSOappend.xyz` (all 10 atoms although types 3 and 4 are excluded: atoms of the central molecule are always written).

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).

[2] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.